Proper Base Change, in one page of dependencies

assembled from: Chow's lemma · PBC⇔IA · Leray · curve calculations · reduction to $\mathbf{P}^1$

Abstract

The proof of the proper base change theorem (PBC) is an assembly of exactly five theorems:

Chow's lemma · PBC⇔IA · Leray · curve calculations (PBC for $\mathbf{P}^1$) · reduction to $\mathbf{P}^1$.

No other statement is introduced: every elementary fact (finite morphisms, composition and cancellation, the symmetric product, étale locality, the degree-zero exchange) is argued *inside* the proofs where it is used. In accordance with `blueprint/src/version/sketch/readme.txt` the *final theorem — proper base change — is written last*: the document builds the ingredients first and then states the target, so that decomposition proceeds backwards from the last theorem. Format conventions of the readme are applied: statements list their dependencies (definitions) in `\uses` with hyperlinks to labels, proofs list what they use and cite it with `\Cref`, `\Cref` and `\uses` agree, and decomposed items are marked `% Decomposed.` The curve-estimate theorem carries a **split marker**: a blue label (pending, per the readme convention) stating explicitly that the size of its decomposition is carried by its *curve-calculations* side — not by the reduction step. Its proof is kept untouched.

1 Definitions

Definition 1.1 (Scheme and morphism conventions)**.** A morphism $f : X \to Y$ is *proper* if it is separated, of finite type and universally closed; *finite* if it is affine and $f_*\mathcal{O}_X$ is a finite $\mathcal{O}_Y$-algebra; an *immersion* is a composite of an open and a closed immersion; *surjective* if surjective on underlying spaces; étale if locally of finite presentation, flat and unramified; and a scheme is *affine* if it is Spec of a commutative ring.

Definition 1.2 (Exchange morphism)**.** *Uses:* `def:conv`

For a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

(i.e. $X' = X \times_Y Y'$) there is a canonical *exchange morphism*

$$\mathrm{Ex}_{\square,F} \;:\; g^{-1}Rf_*F \;\longrightarrow\; Rf'_*g'^{-1}F,$$

the adjoint mate of the counit of $u^{-1} \dashv Ru_*$; it is natural in F and independent of all choices.

Definition 1.3 (The PBC property)**.** *Uses:* `def:conv`, `def:exchange`

For a proper morphism $f : X \to Y$, write $\mathrm{PBC}(f)$ if the exchange morphism $\mathrm{Ex}_{\square,F}$ is an isomorphism in D^+ for every base change and every torsion sheaf F on $X_{\text{ét}}$.

Definition 1.4 (Injective acyclicity)**.** *Uses:* `def:conv`, `def:exchange`

For a proper morphism $f : X \to Y$ and a prime r, write $\mathrm{IA}_r(f)$ if after every base change every injective sheaf I of $\mathbf{F}_r$-modules on $X_{\text{ét}}$ satisfies

$$R^q f'_*\, g'^{-1} I = 0 \qquad (q > 0),$$

and $\mathrm{IA}(f)$ for all primes.

2 The five theorems

2.1 Chow's lemma

Theorem 2.1 (Chow). *Uses:* `def:conv`

Let $f : X \to Y$ be proper with Y affine. There exist a proper surjection $\pi : \widetilde{X} \to X$, an integer n, and a closed immersion $i : \widetilde{X} \hookrightarrow \mathbf{P}^n_Y$ with $f \circ \pi = p \circ i$, where $p : \mathbf{P}^n_Y \to Y$.

Idea. *Uses:* `def:conv`

Cover X by finitely many affine opens $U_1, \dots, U_m$ all of which contain the generic points of all irreducible components, so $U = \bigcap_i U_i$ is dense; embed the U_i in projective spaces, take the diagonal model $\widetilde{X}$ (the closure of U in $\prod_i Z_i$, Z_i the closure of U_i in $\mathbf{P}^m_Y$), glue the projections $p_i : \widetilde{X} \to Z_i$ into $\pi : \widetilde{X} \to X$ (they agree on the dense U); π is proper and surjective. The Segre composite $\widetilde{X} \hookrightarrow \prod_i \mathbf{P}^m_Y \hookrightarrow \mathbf{P}^n_Y$ is a closed immersion, because $\widetilde{X}$ is proper over Y and a proper immersion in the sense of 1.1 Scheme and morphism conventions is closed. □

2.2 The equivalence PBC⇔IA

Theorem 2.2 (PBC ⇔ IA). *Uses:* `def:conv`, `def:pbc`, `def:ia`

For a proper morphism f: PBC(f) *in the sense of 1.3 The PBC property holds if and only if* IA(f) *in the sense of 1.4 Injective acyclicity holds.*

Idea. *Uses:* `def:ia`, `def:exchange`, `thm:leray`

⇒ is immediate: $Rf_*I = f_*I$ for I injective, so the IA_r vanishings of 1.4 Injective acyclicity follow from PBC. ⇐: take an injective resolution $E \to I^\bullet$ of an $\mathbf{F}_r$-sheaf E; 1.4 Injective acyclicity says every $g'^{-1}I^m$ is f'_*-acyclic, so by the criterion of 2.3 Leray(a) the complex $f'_*g'^{-1}I^\bullet$ computes $Rf'_*g'^{-1}E$, while g^{-1} exact makes $g^{-1}f_*I^\bullet$ compute $g^{-1}Rf_*E$. The two complexes are linked termwise by the *underived* exchange of 1.2 Exchange morphism, which is an isomorphism in every degree — and here is the only non-formal step: at a geometric point of Y', both stalks compute sections over the same closed geometric fibre $X_{\bar{s}} = X'_{\bar{t}}$, by the degree-zero comparison $\Gamma(Z, \cdot) \cong \Gamma(Z_s, \cdot)$ for Z proper over a strictly henselian base (Stein's connectedness + restriction on spectral spaces, checked stalkwise, and geometric stalks are conservative). Prime coefficients give all n-torsion sheaves by the exact sequence $0 \to F[r] \to F \to rF \to 0$ and the five lemma; all torsion sheaves are filtered unions $F = \varinjlim F[n]$, over which the proper $R^q f_*$ commutes. □

2.3 Leray

Theorem 2.3 (Leray). *(a)* (Acyclicity criterion.) *If $F \to I^\bullet$ is a resolution by v-acyclic sheaves and v is left exact, then $\mathcal{H}^q(vI^\bullet) \xrightarrow{\sim} R^q vF$: the cohomology of a termwise-acyclic complex computes the derived functor.*

(b) (Spectral sequence for a composite.) *For $X \xrightarrow{f} Y \xrightarrow{h} Z$ there is a functorial spectral sequence*

$$E_2^{p,q} = R^p h_* R^q f_* F \implies R^{p+q}(hf)_* F,$$

compatible with base change.

Use (a) inside 2.2 PBC ⇔ IA; use (b) in 2.5 Reduction to $\mathbf{P}^1$, where composition and cancellation of PBC properties are performed.

2.4 Curve calculations: PBC for the projective line

Theorem 2.4 (Curve calculations ⇒ PBC($\mathbf{P}^1$)). *Uses:* def:conv, def:pbc, def:ia

SPLIT MARKER — the bulk of the decomposition sits on the curve-calculations side of this statement (the strictly-henselian comparison and the separably-closed invariance for proper curves): it may split into a very large number of statements, with its own obligations; the reduction step, in contrast, carries no such size.

For every scheme S, the projection $\mathbf{P}^1_S \to S$ satisfies PBC.

Idea. *Uses:* def:ia, thm:iff

It suffices (2.2 PBC ⇔ IA) to prove IA_r for the projection, i.e. the vanishings of 1.4 Injective acyclicity. The curve calculations enter twice.

- *Strictly henselian base.* For A strictly henselian with closed point s, the degree-zero comparison $\Gamma(Z,F) \cong \Gamma(Z_s, F|_{Z_s})$ (Z proper over A; Stein's connectedness), applied through the Mayer–Vietoris sequence to the two affine opens of $\mathbf{P}^1_A$, kills H^q of an injective for $q \geq 2$; the remaining class in H^1 is killed by the constructible/finite-étale machinery (Gabber's affine comparison + torsor lifting over a proper henselian pair). Hence $H^q(\mathbf{P}^1_A, F) \cong H^q(\mathbf{P}^1_s, F|)$ for every torsion F.
- *Separably closed extension.* For proper curves ($\dim \leq 1$) and $k \subset k'$ separably closed, $H^q(C,F) \cong H^q(C_{k'}, F)$: pass to algebraic closures (universal homeomorphisms leave the étale site unchanged), use constructible reduction, normalization, and the smooth-curve calculations for $\mathbf{F}_r/\mathbf{F}_p$-coefficients.

Now let S be arbitrary. The stalk of the exchange morphism at a geometric point $\bar{t} \to T$ (over $\bar{s} \to S$) is the canonical pullback $H^q(\mathbf{P}^1_{\kappa(\bar{s})}, F) \to H^q(\mathbf{P}^1_{\kappa(\bar{t})}, F)$ between separably closed fields, an isomorphism by the second item. So the exchange map is an isomorphism at every geometric stalk: IA_r holds for the projection (geometric stalks are conservative), and 2.2 PBC ⇔ IA converts it into PBC. □

2.5 Reduction to the projective line

Theorem 2.5 (Reduction to $\mathbf{P}^1$). *Uses:* def:conv, def:pbc

Assume that $\mathbf{P}^1_S \to S$ satisfies PBC *for every scheme S (in the sense of 1.3 The PBC property). Then every proper morphism of schemes satisfies* PBC.

Proof. *Uses:* thm:chow, thm:iff, thm:leray, thm:p1

We use the following elementary manipulations, each argued where it occurs: finite morphisms satisfy PBC; PBC composes; PBC cancels along a proper surjection; $\mathbf{P}^1$ gives $\mathbf{P}^n$; PBC is étale local on the target.

Reduction to an affine base. For quasi-compact quasi-separated f, the restriction of $R^q f_* F$ to an étale $V \to Y$ is the sheafification of $U \mapsto H^q(X \times_Y U, F|)$, compatible with exchange; exchange is therefore an isomorphism iff it is one over the geometric points of an étale cover of Y, and geometric stalks are conservative: PBC is étale local on the target, so we may assume Y affine.

Chow's factorization. By 2.1 Chow there are $\pi : \widetilde{X} \to X$ proper surjective and a closed immersion $i_Y : \widetilde{X} \hookrightarrow \mathbf{P}^n_Y$ with $f\pi = q_Y i_Y$; the graph map $i_X = (\pi, i_Y) : \widetilde{X} \hookrightarrow \mathbf{P}^n_X$ is a closed immersion over X, so $\pi = q_X i_X$.

Closed immersions are finite, and finite morphisms satisfy PBC. After base change to the strict henselization at a geometric point $\bar{y}$ the finite fibre is a finite disjoint union of strictly henselian local schemes, whence $(f_*F)_{\bar{y}} = \prod_{\bar{x} \to \bar{y}} F_{\bar{x}}$ as a finite product; f_* is exact, $R^q f_* = 0$ for $q > 0$, exchange is the identity summand-wise, and the same holds after every base change. Thus $\mathrm{PBC}(i_X) = \mathrm{PBC}(i_Y)$.

From $\mathbf{P}^1$ *to* $\mathbf{P}^n$. The symmetric-product morphism

$$\sigma_n : (\mathbf{P}^1_S)^n \to \mathbf{P}^n_S, \quad ([x_i : y_i])_i \mapsto [F_0 : \cdots : F_n], \quad \prod_{i=1}^n (x_i t + y_i) = \sum_j F_j t^j,$$

is finite (each $z_i = x_i/y_i$ is integral over the elementary symmetric subring) and surjective (over an algebraically closed field every binary form of degree n factors into linear forms). The projection $(\mathbf{P}^1_T)^n \to T$ is a composite of n projective-line projections over successive bases: by 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$) and the composition step below it satisfies PBC; σ_n being finite satisfies PBC; and since the composite $(\mathbf{P}^1_T)^n \to \mathbf{P}^n_T \to T$ is the product projection, cancellation along the surjective σ_n (below) gives PBC for $\mathbf{P}^n_T \to T$.

Composition. If f, h are proper with PBC(f), PBC(h), then PBC(hf): convert both to IA via 2.2 PBC $\Leftrightarrow$ IA; $f_* I$ is injective for I injective (f_* is right adjoint to the exact f^{-1}), the $q > 0$ higher direct images of $a^{-1}I$ along f' vanish, and the conjugates are identified by the degree-zero exchange used in 2.2 PBC $\Leftrightarrow$ IA; in the spectral sequence 2.3 Leray(b) for the composite only the $q = 0$ row survives, killed in positive degrees by IA(h); convert back. Hence PBC(π) ($\pi = q_X i_X$) and PBC($f\pi$) ($f\pi = q_Y i_Y$).

Cancellation. If f is proper surjective with PBC(f), PBC(hf), then PBC(h): convert to IA; an injective I on Y embeds $I_{\bar{y}} \hookrightarrow J_{\bar{x}}$ into an injective J on X over the fibre, so $I \to f_* J$ splits and I is a direct summand of $f_* J$; by the spectral sequence 2.3 Leray(b) together with IA(f) and IA(hf) the higher direct images of $b^{-1} f_* J \cong f'_* a^{-1} J$ vanish, and vanishing passes to direct summands; convert back. 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$) supplies the hypothesis, so cancel π: PBC(f). □

3 The final theorem

Theorem 3.1 (Proper base change)**.** *Uses:* `def:conv`, `def:pbc`, `def:exchange`

For a proper morphism $f : X \to Y$ and every torsion sheaf F on $X_{\text{ét}}$, PBC(f) holds in the sense of 1.3 The PBC property: the exchange morphism

$$\mathrm{Ex}_{\square,F} : g^{-1} R f_* F \xrightarrow{\sim} R f'_* g'^{-1} F$$

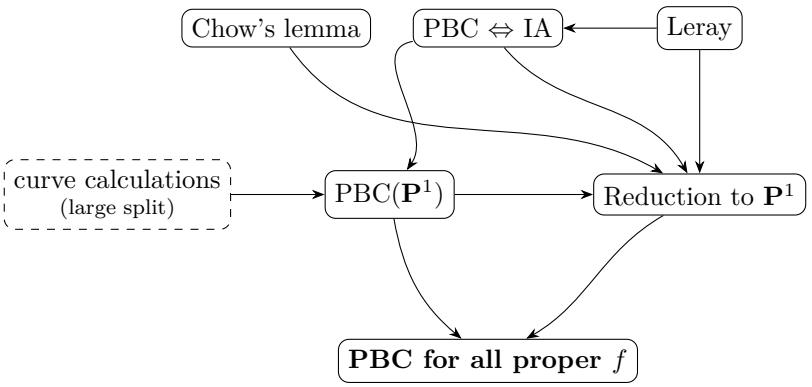
is an isomorphism.

Proof. *Uses:* `thm:p1`, `thm:reduction`

Step 1 ($\mathbf{P}^1$). By 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$), $\mathbf{P}^1_S \to S$ satisfies PBC for every S.

Step 2 (reduction). 2.5 Reduction to $\mathbf{P}^1$ now applies: every proper morphism f satisfies PBC(f) for $\mathbf{F}_r$-coefficients, all primes r.

Step 3 (finite torsion). A sheaf killed by n is decomposed by the exact sequences $0 \to F[r] \to F \to rF \to 0$ (induction on the number of prime factors of n, five lemma); any torsion sheaf is the filtered union of its n-torsion parts, over which the proper $R^q f_*$ commutes. Hence PBC(f) for every torsion sheaf: 3.1 Proper base change. □



Dependency graph (arrows = `\Cref`-dependencies, i.e. the *proof*-level `\uses` fields):
2.5 Reduction consumes 2.1 Chow, 2.2 PBC $\Leftrightarrow$ IA, 2.3 Leray(b) and 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$);
2.4 PBC($\mathbf{P}^1$) consumes 2.2 PBC $\Leftrightarrow$ IA; 2.2 PBC$\Leftrightarrow$IA consumes 2.3 Leray(a);
the final theorem consumes 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$) (Step 1) and 2.5 Reduction to $\mathbf{P}^1$ (Step 2).
The horizontal arrow PBC($\mathbf{P}^1$) $\to$ Reduction is the theorem's *hypothesis* input: the proof of 2.5 cites 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$) both for $\mathbf{P}^n$ and to supply the hypothesis ("2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$) supplies the hypothesis"); all other arrows into Reduction feed its internal argument.

Remark 3.2. Only the five theorems of §2 are statements; every other fact used in the proofs is argued inline (finite morphisms, composition, cancellation, the symmetric product, étale locality, the degree-zero exchange). The curve-estimate theorem 2.4 Curve calculations $\Rightarrow$ PBC($\mathbf{P}^1$) is the single step carrying the split marker: it alone sits on the critical path (Step 1 of the final theorem and the hypothesis of the reduction), and its own decomposition is by far the largest one. The target theorem is last, as the readme's decomposition order prescribes. The blueprint labels: `thm:weil-pbc-chow`, `prop:weil-pbc-property-iff-acyclicity`, `lem:weil-pbc-leray-acyclicity`, `prop:weil-pbc-projective-line`, `prop:weil-pbc-reduction-to-projective-line`.